

Homework 10

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Problem 1

Let (X, d) be a metric space and let $Y \subseteq X$. We call the set Y complete if the subspace Y with the induced metric is complete. Prove that if Y is complete, then Y is closed in X . (Hint: It suffices to show $\overline{Y} \subseteq Y$. Use sequences.)

Discussion & Scratch Work

Proof.

□

Problem 2

Let (X, d) be a metric space. Prove that if $A, B \subseteq X$ are both compact then $A \cup B$ is compact.

Discussion & Scratch Work

Proof.

□

Problem 3

Let (X, d) be a metric space.

- (a) Prove that if $A \subseteq B \subseteq X$, A is closed in X , and B is compact, then A is compact.
(Hint: Given an open cover of A , we can add A^c to get an open cover of B .)
- (b) Prove that if $C, D \subseteq X$ are both compact, then $C \cap D$ is compact. (Hint: Use (a).)

Discussion & Scratch Work

Proof.

□

Problem 4

Let (X, d) be a metric space and let $Y \subseteq X$. Prove that if Y is compact, then Y is complete.

Discussion & Scratch Work

Proof.

□

Appendix: Toolbox (optional)